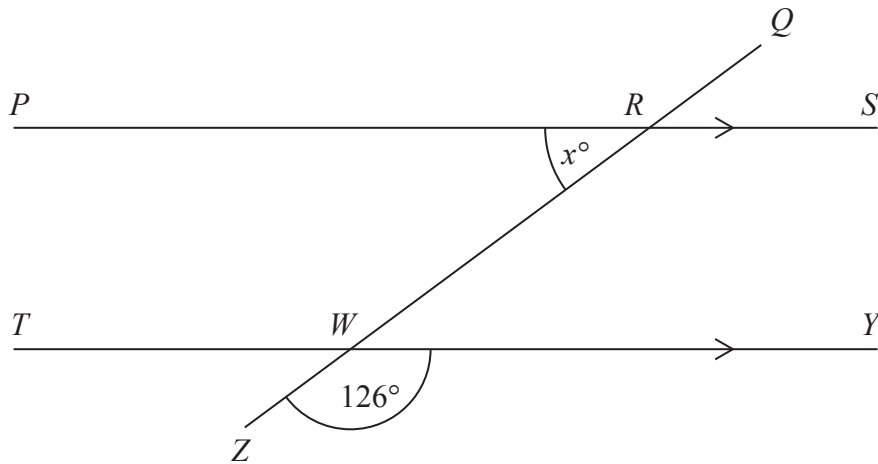


1

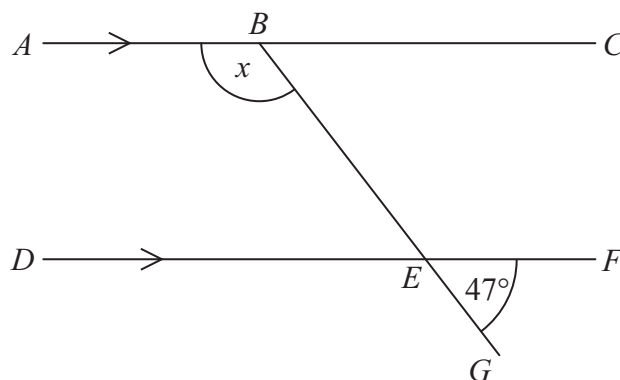


PRS and TWY are parallel straight lines.
 $QRWZ$ is a straight line.

Work out the value of x .
Give reasons for your answer.

(Total for Question 1 is 3 marks)

2



ABC and DEF are parallel lines.

BEG is a straight line.

Angle $GEF = 47^\circ$.

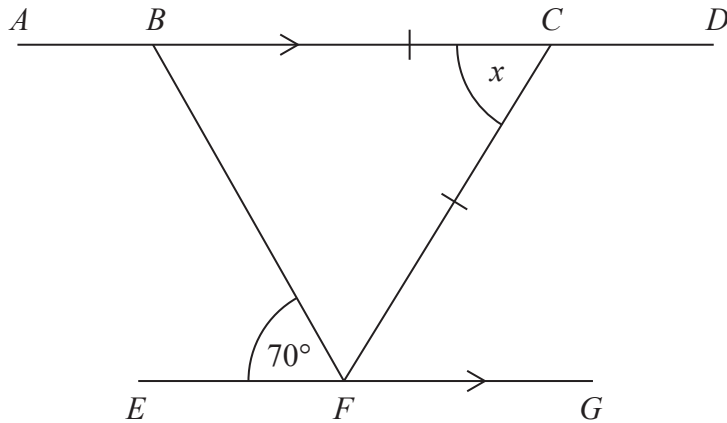
Work out the size of the angle marked x .

Give reasons for your answer.

o

(Total for Question 2 is 3 marks)

3



$ABCD$ and EFG are parallel lines.

$BC = CF$

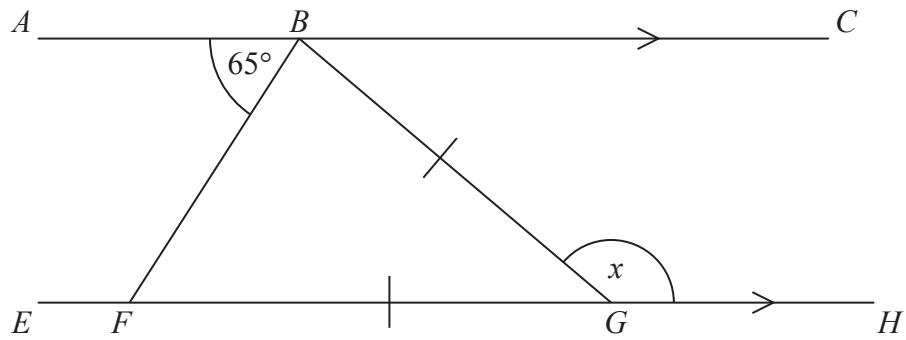
Angle $BFE = 70^\circ$

Work out the size of the angle marked x .

Give reasons for each stage of your working.

(Total for Question 3 is 4 marks)

4



ABC is parallel to $EFGH$.

$$GB = GF$$

$$\text{Angle } ABF = 65^\circ$$

Work out the size of the angle marked x .

Give reasons for your answer.

(Total for Question 4 is 4 marks)

5

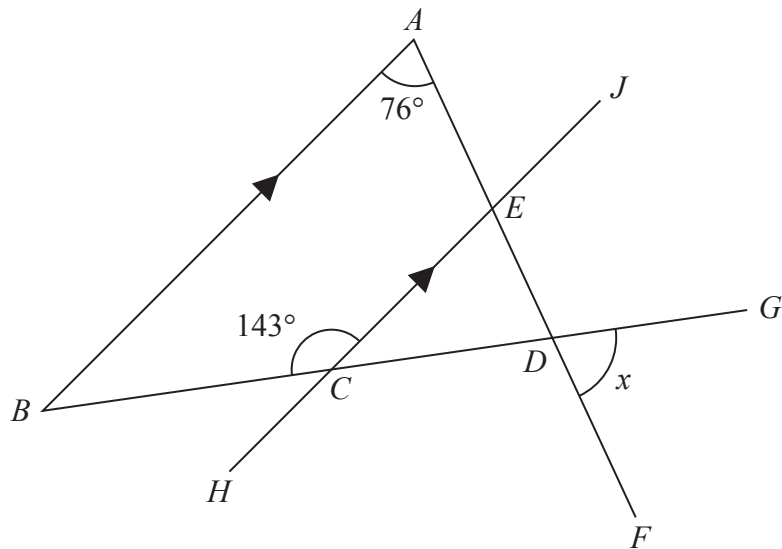


Diagram **NOT**
accurately drawn

ABD is a triangle.

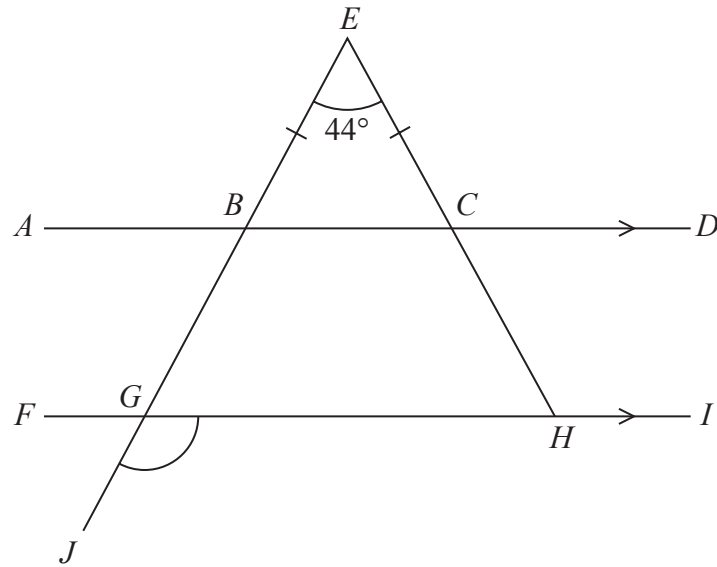
$AEDF$, $BCDG$ and $HCEJ$ are straight lines.

BA is parallel to $HCEJ$.

Work out the size of the angle marked x .

(Total for Question 5 is 3 marks)

6



$ABCD$ and $FGHI$ are parallel straight lines.
 $EBGJ$ and ECH are straight lines.

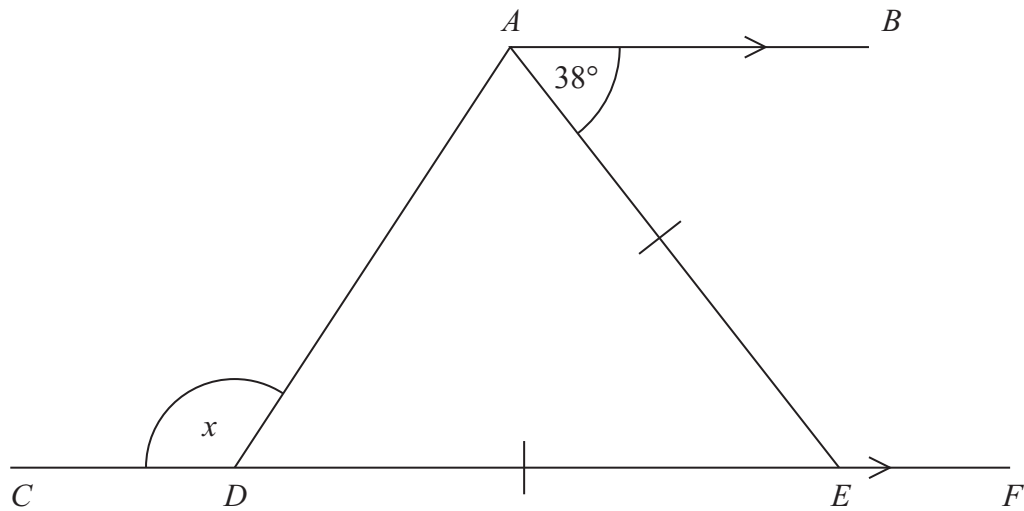
$$BE = CE$$

$$\text{Angle } BEC = 44^\circ$$

Work out the size of angle JGH .
 Give a reason for each stage of your working.

(Total for Question 6 is 5 marks)

7



$CDEF$ is a straight line.

AB is parallel to CF .

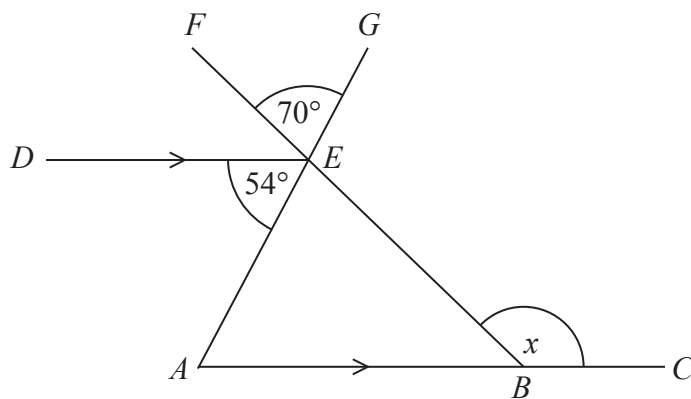
$DE = AE$.

Work out the size of the angle marked x .

You must give reasons for your answer.

(Total for Question 7 is 4 marks)

8



ABC and DE are parallel lines.
 AEG and BEF are straight lines.

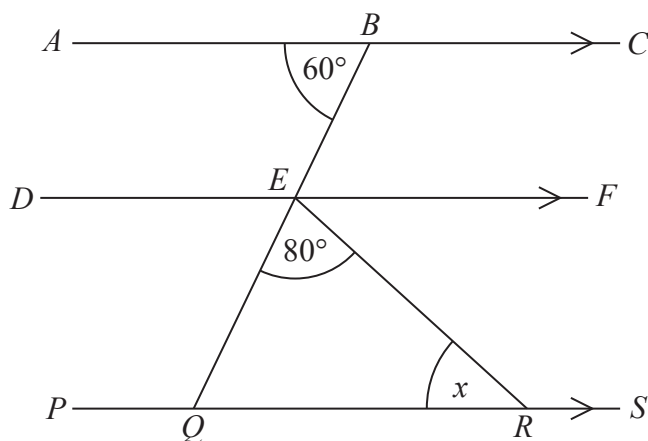
Angle $AED = 54^\circ$

Angle $FEG = 70^\circ$

Work out the size of the angle marked x .
 Give a reason for each stage of your working.

(Total for Question 8 is 4 marks)

9



ABC , DEF and $PQRS$ are parallel lines.
 BEQ is a straight line.

Angle $ABE = 60^\circ$

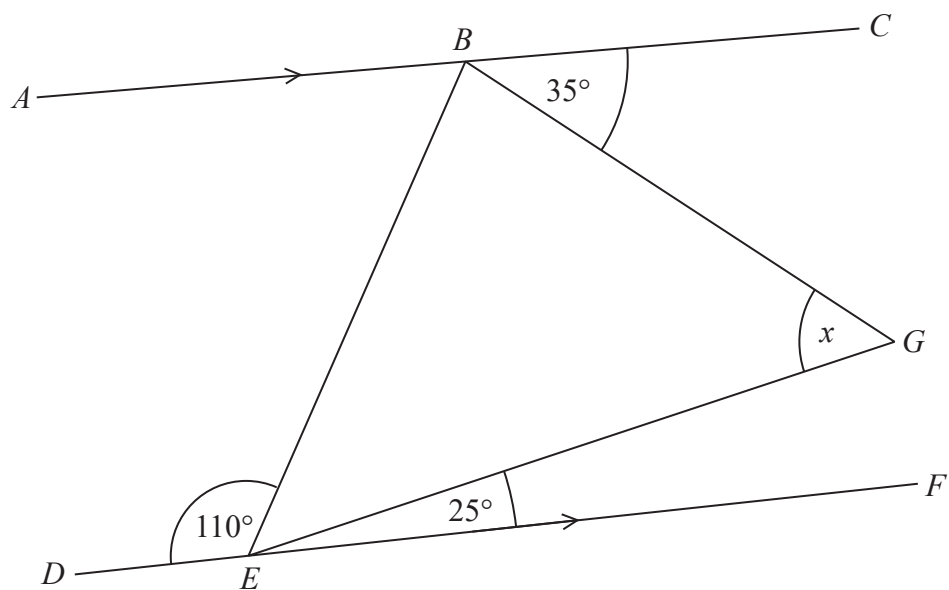
Angle $QER = 80^\circ$

Work out the size of the angle marked x .

Give reasons for each stage of your working.

(Total for Question 9 is 4 marks)

10 BEG is a triangle.



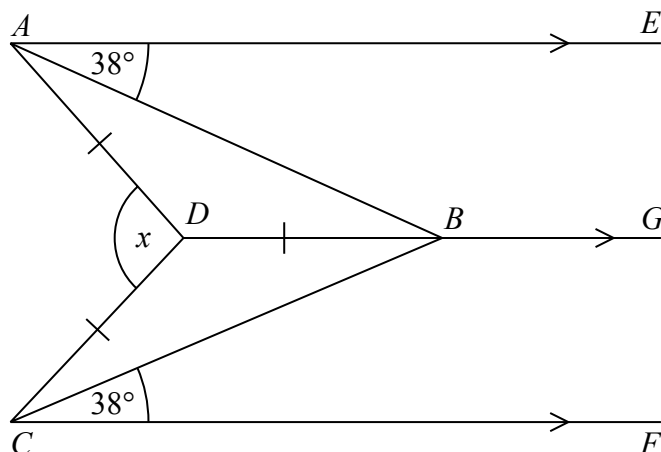
ABC and DEF are parallel lines.

Work out the size of angle x .

Give a reason for each stage of your working.

(Total for Question 10 is 4 marks)

11



AE , DBG and CF are parallel.

$DA = DB = DC$.

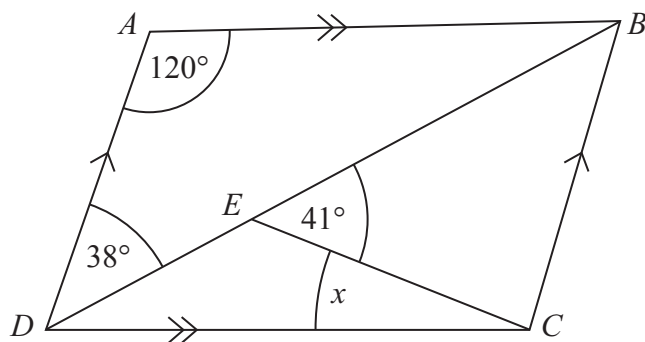
Angle $EAB = \text{angle } BCF = 38^\circ$

Work out the size of the angle marked x .

You must show your working.

(Total for Question 11 is 3 marks)

12



$ABCD$ is a parallelogram.

Angle $ADB = 38^\circ$.

Angle $BEC = 41^\circ$.

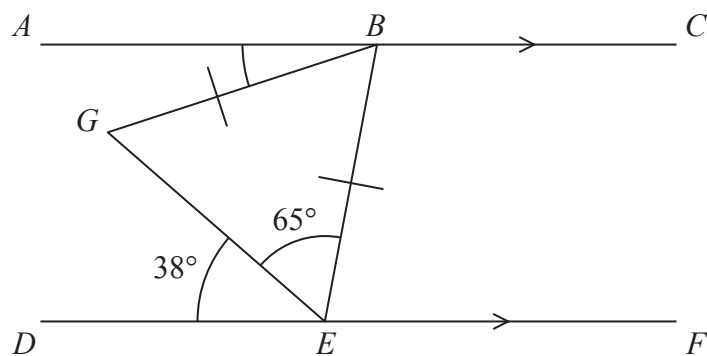
Angle $DAB = 120^\circ$.

Calculate the size of angle x .

You must give reasons for your answer.

(Total for Question 12 is 4 marks)

13



ABC and DEF are parallel lines.

$BG = BE$

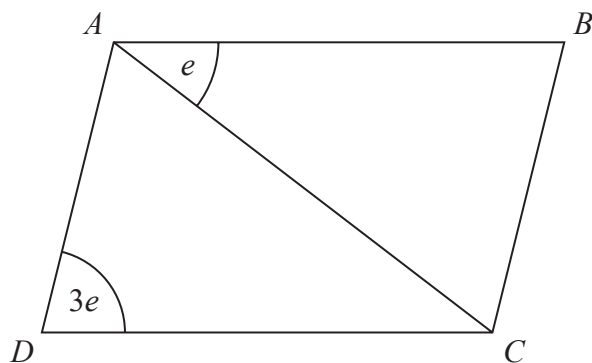
Angle $DEG = 38^\circ$

Angle $GEB = 65^\circ$

Find the size of angle ABG .

(Total for Question 13 is 3 marks)

14 $ABCD$ is a parallelogram.



All angles are measured in degrees.

Find an expression, in terms of e , for the size of angle CAD .

Give a reason for each stage of your working.

(Total for Question 14 is 3 marks)

15

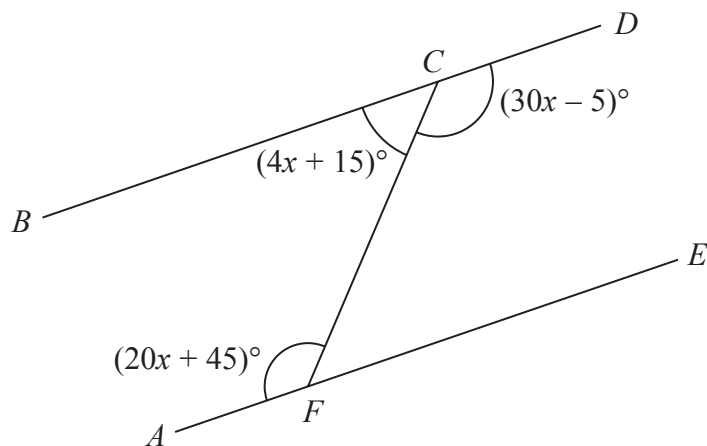


Diagram **NOT**
accurately drawn

BCD and AFE are straight lines.

Show that BCD is parallel to AFE .

Give reasons for your working.

(Total for Question 15 is 5 marks)